

Advances in Multi-Style and Real-Time Transfer

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Abstract—This paper provides an overview of neural style transfer techniques, focusing on multi-style and real-time applications for both images and videos. Multi-style transfer refers to the technique of combining several art styles based on a given content image to create a unique and intriguing visual result. In contrast, real-time style transfer involves applying artistic styles almost instantaneously using pre-trained feedforward networks, making it particularly well-suited for video editing and live broadcasts. This paper reviews the most significant research and technical advancements in the field, while also addressing existing challenges that hinder the full potential of these transformative technologies. Furthermore, attention is given to emerging applications and the areas still in need of improvement, to enhance understanding and practical implementation.

Index Terms—Multi-Style Transfer, Real-time Style Transfer, Image Style Transfer, Video Style Transfer

I. INTRODUCTION

Multi-style transfer is one of the higher functionalities of neural style transfer, where several styles are applied to one base image. It takes in one content image and multiple style images, extracting features of the content image through CNNs and applying styles learned from different style images as a final output. Real-time style transfer is a disruptive innovation that applies artistic styles to images or videos rapidly. Without using any iterative optimizations, it relies on an instantaneous feed-forward neural network that has been pre-trained to apply styles immediately. These are ideal for immediate feedback applications such as video editing and live streaming. Techniques covered in this paper include the following:

1. Neural Algorithm of Artistic Style: Gatys et al. [1] proposed that image content could be separated from artistic style using convolutional neural networks and then combined by mixing the content of one image with the style of another.
2. Perceptual Losses for Real-Time Style Transfer and Super-Resolution - Johnson et al. [2] approach makes use of large, pre-trained networks through the perceptual loss functions to carry out real-time style transfer and super-resolution. These results are of higher speed and quality with great content and style preservation.

3. Fast Video Multi-Style Transfer: Wang et al. [3] framework was developed for online video multi-style transfer. It leverages multiple instance normalizations, along with ConvLSTM modules, to preserve the continuity between frames and minimize flickering effects.

4. Multi-Style Generative Network for Real-Time Transfer, MSG-Net: Zhang et al. [4] MSG-Net utilizes CoMatch Layer in capturing the second order style statistics, allows for fast processing, fine control over the qualities of images in a very efficient model size while keeping colors and the size of a brush. Motivation for the Approach - The recent progress in the transfer of neural style supports the creative forms of digital media more highly. This paper seeks to focus on innovations and challenges by investigating multi-style as well as real-time applications. As a result, instigated further development.

This study focuses on the innovations and challenges as shown in Figure 1. It seeks to understand how real-time and multi-style transfers function, their evolution, key features, and the challenges they face, particularly in image and video processing. Additionally, the study explores current and future trends in this field. This paper is organized as follows: Section II discusses the history, working principles, and challenges of neural style transfer techniques. Section III presents current trends, advantages, disadvantages, and future directions in the field. Section IV offers a final discussion, while Section V concludes the paper.

II. HISTORY, WORKING, AND CHALLENGES

Before neural networks, various texture synthesis techniques were used for style transfer. Litwinowicz [5] and Haeberli [6] pioneered impressionist effects and abstract image representations, laying the groundwork for stylistic manipulation of visual content. Efros and Leung [7] advanced texture synthesis with a non-parametric sampling method, generating new textures by replicating patches from existing images. [8] further improved texture quality with 'Image Quilting' for better texture transfer. Image Analogies by Hertzmann

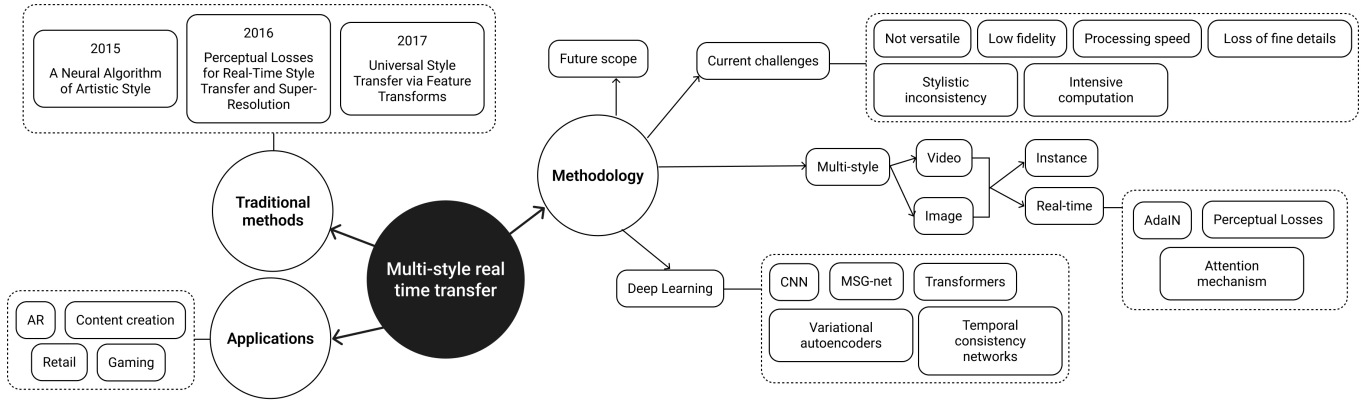


Fig. 1. Concept Map of Multi-Style and Real-Time Transfer: This diagram illustrates the evolution of techniques, highlighting Traditional Methods, Deep Learning Approaches, Applications, Current Challenges, and Future Scope in Image and Video Style Transfer.

et al. [9] introduced a method for learning mappings from example image pairs, enabling stylistic effects and transformations based on observed patterns. These non-parametric methods, based on low-level features, faced challenges like global inconsistency and limited flexibility. Neural networks, especially CNNs, revolutionized style transfer by capturing semantic content and rendering texture from a reference style. Gatys et al. [1] first pioneered this approach by merging their parametric texture synthesis [10] method with the image representation inversion techniques [11]. They used features from different CNN (VGG) layers to extract content (higher layers) and style (lower layers) and introduced three key loss functions:

Content loss: Measures the difference between feature maps of the generated and content images, computed as the mean squared error between activations at a specific layer.

$$L_{\text{content}} = \frac{1}{2} \sum_{i,j} (F_{i,j}^G - F_{i,j}^C)^2 \quad (1)$$

where $F_{i,j}^G$ and $F_{i,j}^C$ are activations at the (i, j) -th positions in the generated and content image feature maps [1].

Style loss: Measures the difference between the Gram matrices of the generated and style images. Gram matrix captures correlations between filter responses within a layer, and comparing these matrices helps ensure the generated image matches the style's texture and pattern.

$$L_{\text{style}} = \sum_{W_i} \frac{1}{4N_i^2 M_i^2} \sum_{i,j} (G_{i,j}^G - G_{i,j}^S)^2 \quad (2)$$

where G^G and G^S are the Gram matrices for the generated and style images, and N_i , M_i , and W_i are the dimensions and weights of the feature maps at layer i [1].

Total loss: It is a weighted sum of content and style losses, with factors balancing their contributions. The loss is minimized iteratively using optimizers like LBFGS or Adam to refine the generated image from random noise. This process is computationally intensive, making real-time applications challenging.

To address this, Johnson et. al. [2] introduced a feed-forward neural network trained with perceptual, style, and content losses, enabling real-time style transfer. They optimize the weights of an image transformation network rather than pixel values. Similar to Gatys [1], it uses a combined loss function derived from a deep network to guide the training. The style loss captures perceptual similarity more effectively than traditional pixel-wise methods. For each image, the style loss gets fixed while the content loss varies, allowing the network to apply style consistently across different content images. By doing so the network learns to minimize style on arbitrary images. It achieves results comparable to Gatys' LBFGS method but is significantly faster. However, it supports only one style per network, making it less flexible than methods that can handle multiple styles without retraining. Building on this work, [12] introduced Instance Normalization in place of Batch Normalization for fast stylization. It normalizes each image independently, enhancing style transfer with greater flexibility by improving style and content separation. Subsequent research has explored variations and improvements to Instance Normalization for broader applications. Research has improved Instance Normalization with Conditional Instance Normalization (CIN), which uses conditional parameters for better style blending [13]. CIN applies two affine parameters (scaling and shifting) to define new styles while keeping convolution parameters consistent.

Adaptive Instance Normalization (AdaIN) enhances neural style transfer by aligning neural activation distributions between content and style images. [14] used a pre-trained encoder-decoder network to extract features from both the content and style images. They applied instance normalization to the content features, adjusting their mean and variance separately for each image. AdaIN was then used to adapt these normalized content features to match the mean and variance of the style features. Importantly, the affine parameters γ and β used in traditional normalization techniques do not require learning during the process. Instead, it adapts content features to style features using the style image's mean and variance,

enabling real-time transfer. However, the decoder must be trained on a finite set of style images to reconstruct the stylized image effectively.

$$\text{AdaIN}(x, \gamma, \beta) = \gamma \cdot \left(\frac{x - \mu_x}{\sigma_x} \right) + \beta \quad (3)$$

where x represents the normalized content features, μ_x and σ_x are their mean and variance, and γ and β denote the mean and variance of the style features [14].

Gatys et al. [15] advanced control over perceptual factors, focusing on spatial characteristics, color, and scale. They introduced guided Gram Matrices for applying different styles to specific image regions, preventing blending and preserving style coherence. Techniques like histogram matching and luminance transfer ensure that color effects are maintained even with multiple styles applied. The previous feed-forward generative models often struggled with image quality and flexibility, especially when handling multiple or arbitrary styles. Zhang and Dana's Multi-Style Generative Network (MSG-Net) [4] introduced a novel approach to real-time style transfer designed for both efficiency and flexibility. MSG-Net introduces the CoMatch Layer, a novel component that matches the second-order feature statistics, specifically the Gram Matrix, of target styles. This provides a richer representation of style compared to 1-dimensional embeddings. It employs an upsampled convolution strategy to prevent checkerboard artifacts, a common issue with fractionally-strided convolutions. However, MSG-Net requires extensive pre-training with a diverse set of style images, limiting its adaptability to new styles without retraining.

Maintaining temporal coherence in video processing is crucial. Independent frame processing can lead to flickering and inconsistencies, as each frame might converge to different local optima [16]. Penalizing differences between consecutive frames can ensure smooth transitions but slows down the process, as each frame must be adjusted based on its predecessor. To address these issues, [17] proposed a Real-Time method using a feed-forward CNN that ensures temporal consistency. The approach involves a stylizing network for individual frames and a loss network that computes spatial and temporal losses. Spatial loss measures alignment with the style reference image, while temporal loss uses pre-computed optical flows to ensure continuity between frames. Despite this, the method struggles with long-term temporal coherence and capturing low-level details, resulting in reduced clarity in strokes and textures due to the lower emphasis on perceptual losses. [18] proposed ReCoNet addresses these issues maintaining both high perceptual quality and temporal consistency. ReCoNet consists of an encoder-decoder, which transforms each input frame into a stylized output. For temporal coherence, it introduces a novel feature-map-level temporal loss and a luminance warping constraint, addressing issues of illumination changes and flickering. The real-time arbitrary video style transfer method in [19] used a three-network setup: prediction, stylization, and loss. The prediction network extracted style parameters, which the stylization network applied to each frame.

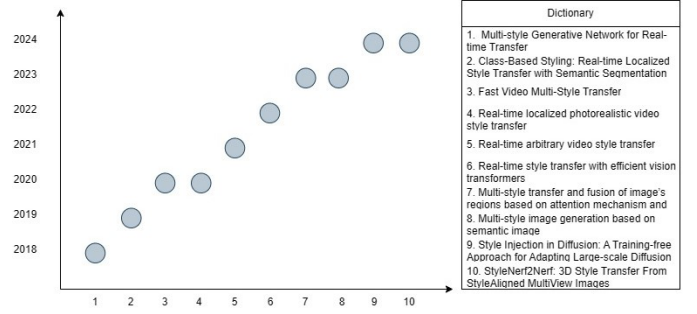


Fig. 2. The timeline of trends illustrates the progression of current developments in the field since 2018.

The loss network optimized these parameters by integrating content, style, and temporal consistency losses. The training involved using adjacent frames and the style image to compute losses and ensure temporal coherence. Once trained, the stylization network processed frames independently, enabling real-time performance. [3] introduced a framework for applying up to 120 artistic styles to video content with a single network, featuring a Multi-Instance Normalization Block and ConvLSTM modules for temporal coherence. This method ensures diverse styles and smooth, flicker-free outputs. [20] proposed a Real-time Localized Photo-realistic Video Style Transfer method using a dual-path architecture: a low-resolution grid prediction path for encoding style transformations and a high-resolution full-frame processing path for rendering. Unlike previous methods that blend styles at the pixel level, this technique blends styles in a lower-dimensional grid space, allowing for smoother transitions and reduced artifacts. It addresses common issues such as mask imperfections and temporal flickering by incorporating a mask enhancement network and a spatiotemporal AdaIN layer, ensuring temporal coherence and consistent style application. Recent real-time style transfer advancements utilize transformers. One study optimizes a transformer-based model for mobile devices using Neural Architecture Search to enhance performance [21]. Another approach, StyleFormer, integrates transformers with a style bank and parametric composition, improving efficiency and visual quality while preserving content structure [22].

III. CURRENT TRENDS, DISADVANTAGES, ADVANTAGES, AND FUTURE DIRECTIONS

A. Current Trends

3D Style Transfer: Recent developments like using Neural Radiance Fields (NeRF) allow us to create 3D assets or 3D scenes from real-world input without much effort. Editing these 3D scenes, especially complex mesh-based ones, is difficult and needs specialized skills. The work by Fujiwara et al. (2024) [23] addresses this by proposing a novel approach that generates stylized images using a text-guided diffusion model along with depth maps from NeRF views. This method allows previewing and adjusting styles before fine-tuning the 3D model with Sliced Wasserstein Distance (SWD) loss. SWD helps to ensure that the stylized 3D scene aligns well with the

desired style. Lui et al. [24] proposed StyleGaussian, which is a new technique for applying artistic styles to 3D scenes quickly. The process includes three main steps: embedding 2D image features into 3D models, applying the style from a reference image given, and then rendering the final scene.

Segmentation-Based Style Transfer: The style is applied only to the background first using the semantic segmentation technique. The foreground is adjusted with the color transfer. The two are blended back in using feathering techniques to avoid artifacts. The blending includes the use of a canny edge detector and then applying a Breadth-first search algorithm to remove points that do not align with the boundary, especially those whose closest edge points lie inside the mask. This is followed by Euclidean distance-based smoothening.

$$C_{\text{blend}} = \alpha \cdot C_{\text{color}} + (1 - \alpha) \cdot C_{\text{style}} \quad (4)$$

Traditional style transfer methods often struggle with maintaining object characteristics during stylization, and multi-stage processes can unnecessarily complicate deployment. The research by Kulkarni et al. [25] introduces a unified approach using YOLOv8 for both object segmentation and style transfer by combining the YOLOv8x-seg segmentation model with YOLOv8's backbone for style transfer, simplifying the architecture. The framework includes three main modules: a segmentation module for separating the objects, a style transfer module, and a blending module that merges the separated objects back into the original image.

Multimodality in Style Transfer: The paper [26] introduces a new method called MultiModality-guided Image Style Transfer (MMIST) to improve the output provided by text-guided methods. This technique allows style instructions from multiple modes, such as both text and images. The key innovation is a technique called cross-modal GAN inversion, which generates style representations from various inputs.

Diffusion Models in Style Transfer: The proposed LSAST framework [27] addresses the issue of generating realistic outputs by introducing a Step-aware and Layer-aware Prompt Space. This dynamically adjusts both the content structure and style patterns through a series of learnable prompts across different steps and layers of the diffusion model. LSAST uses a new inversion technique to train prompts and includes a pre-trained ControlNet branch to better maintain the structure of the content.

Zero-Shot /Training-free Style Transfer: Chung et al. [28] introduce a training-free artistic style transfer method that uses pre-trained large-scale diffusion models. The self-attention features are manipulated by replacing the key and value features of the content image with those obtained from the artistic image. This ensures that the content of the actual image is preserved. Thus, it takes advantage of the similarity-based attention mechanism to improve output quality. It introduces Query preservation which maintains the spatial structure of the original content. Attention temperature scaling sharpens the attention maps. The initial latent AdaIN corrects color inconsistencies in the result.

Transformer-Based Style Transfer: Work done by Zhang et al. [29] focuses on producing high-quality stylized outputs. The work proposes Style Consistency Instance Normalization (SCIN) to better align content and style features. The paper introduces Instance-based Contrastive Learning (ICL) to understand the relationships between styles and reduce artifacts. The Perception Encoder (PE) extracts style features more effectively. Objective loss equation: Where L_{Adv} is adversarial loss, L_{identity} preserves structure, L_c is perceptual content loss, L_s is style loss, and L_{contra} is contrastive loss.

$$L = \lambda_1 L_s + \lambda_2 L_c + \lambda_3 L_{\text{identity}} + \lambda_4 L_{\text{Adv}} + \lambda_5 L_{\text{contra}} \quad (5)$$

B. Disadvantages

The main drawbacks of traditional style transfer methods are that they suffer from a lack of flexibility in adjusting brush stroke sizes in real time and produce poor image quality, limiting their effectiveness. Early approaches used 1-dimensional embeddings to represent artistic styles. This resulted in a lack of details and the generation of visually unappealing results, which counters the purpose of artistic style transfer techniques. MSG-Net mainly focuses on improving image quality. In video style transfer, the main challenge is to maintain temporal cohesion across frames in real-time. This limitation stresses on the need for further innovations, such as those explored by Gao et al. (2020) [3].

Style transfer in 3D reconstructions [23] struggles to work on smaller, delicate details, fine textures, and variation in the background. Depth maps limit the ability to modify the geometry of the scene, and the Gaussian Splatting approach [24] only works on modifying. Thus, styles are not transferred accurately. Multiple 3D objects can also significantly increase memory. In terms of object segmentation [30] [25], the framework's usefulness can be increased by training it to handle a diverse range of objects in the foreground. Segmentation can also be extended to work on 3D scenes. The multi-modal approach [26] currently works with text and images. Future research can be directed towards other modes such as audio (non-speech sounds), speech, video, haptics, etc. Traditional methods based on generative adversarial networks (GANs) often struggle with realistic outputs and can introduce artifacts. Recent advances with large-scale pre-trained diffusion models have shown promise in generating highly realistic images. However, the issue with diffusion-based approaches is that they sometimes fail to preserve the content structure of input images correctly.

C. Advantages

MSG-Net [4] incorporates several key components, including the CoMatch Layer, Upsampled Convolution, and Upsampling Residual Blocks to handle the limitations of prior approaches. By employing feed-forward convolutional neural networks, MSG-Net processes images in a single pass, ensuring high-speed performance. The network builds on the approach proposed by Johnson et al. (2016) [2]. Gao et al. (2020) [3] approach addresses the issue of flickering effects across frames. Prior approaches still suffer in trying to strike

a balance between generalizability and performance. Most of the approaches work well on short-duration clips with a single style. The VMST framework [3] provides a solution by enabling rapid multi-style video transfer within a single model.

Lui et al. [24] proposed StyleGaussian, for applying artistic styles to 3D scenes quickly. It works at 10 frames per second. It uses 3D Gaussian Splatting to achieve this without affecting real-time rendering. The method reduces memory use and introduces a special 3D CNN for consistent multi-view results. It provides high-quality and instant 3D style transfer while maintaining fast performance. [30] introduced a new method for style transfer that focuses on solving the problem of uneven styling in images with prominent foreground objects, like people or other objects. The research by [25] addresses the challenge of integrating style transfer and object segmentation within a single deep neural network. Paper [26] introduces a new method called MultiModality-guided Image Style Transfer (MMIST). This technique helps combine styles from different modalities and creates better-stylized output. The proposed LSAST framework [27] addresses the issues considered with diffusion. [28] introduces a new method for artistic style transfer that uses the power of pre-trained large-scale diffusion models, while eliminating the need for optimization or supervision.

D. Future Scope

Temporal coherence: MSG-Net improves image quality and handles varying brush stroke sizes, but it struggles with maintaining temporal coherence. [3] has specified that even extending MSG-Net by adding temporal loss or ConvLSTM modules, the results were unappealing and temporally inconsistent. Further improvements are needed for coherence.

Number of styles vs real-time: The Multi-style Generative Network (MSG-Net) is limited to 1K styles and the VMST model to 120 styles. Arbitrary style models are designed for diverse inputs. Handling this requires extensive pre-processing mainly for flickering videos. This hurts real-time performance. Future work should focus on more efficient algorithms and architectures to reduce the overhead computation required for adding more styles.

Interactivity and output quality: Creating tools that let users adjust styles in real-time could make things much better and more versatile. Future research should explore advanced methods for achieving more seamless style integration, potentially through improved attention mechanisms and novel blending techniques that ensure a more cohesive final output. Despite advancements, output quality often suffers from issues with detail preservation and resolution. Continued refinement of algorithms is necessary to enhance quality and minimize artifacts.

3D and multi-modality: Extending style transfer methods to 3D models and virtual reality, integrating style transfer different modalities could open new avenues for creative and practical applications. Future research should focus on reduc-

ing memory usage, boosting processing speed, and improving output quality.

IV. FINAL DISCUSSION

Early image style transfer methods using classical image processing tools (e.g., edge detectors, image pyramids) laid an important foundation but lacked controllable parameters and acceptable visual quality. These methods produced subpar results compared to modern neural network-based approaches. Deep learning has since provided a robust framework for significant advancements in style transfer. Gatys et al. [10], [1], illustrated that CNNs could manage complex styles within the confines of an artistic perspective. Although the subsequent deep learning methods were achieved and produced high-quality images, these did not help in real-time application. The research of Johnson et al. [2], took an important technical step forward within real-time applications, using perceptual loss functions to increase the image quality and speed of processing. Huang et al. [14], took this a step further with the introduction of AdaIN, which achieved greater flexibility in theming as well as a short bidirectional computation within processing. Notable, other new models introduced quick computation through variable approaches, such as Zhang et al.'s MSG-Net [4], which used latent style coding and skip connections to achieve exceptional quality along with the speed of processing.

Gao et al. [18] addressed style consistency within the domain of video. As with all factors of neural processing image content, temporally consistent and thematic stable without flicker is one of the more difficult proceeding tasks when processing high-fidelity media. Adequately, in terms of tag, the more recent visual quality which can now be relatively uncertain of further academic field exploration is the idea of image style transfer incorporating, attentional, and understanding the semantics of the local spatial association of the image [15] [22]. Additionally, often lower performance with models rich in mechanisms; this alone will not eliminate but possibly contribute to the need for improvement performance while maintaining numerous styles, images, etc. [16] [3] [19]. Methods like MSG-Net, VMST, and AdaIN outperform traditional approaches in both style fidelity and speed. MSG-Net processes images at 92.7 FPS (256px) compared to Gatys et al.'s 0.07 FPS while delivering superior style fidelity. VMST achieves near real-time video stylization at 19ms per frame (256 X 384px), producing coherent videos with fewer artifacts. AdaIN supports arbitrary styles, processing images in just 0.018s (256px), significantly faster than Gatys et al.'s 14.17s, offering a balance between speed and flexibility better than traditional methods.

Future research will become consolidated toward developing models within a framework in nursing in a cross-platform setting utilizing a wider range of styles and compositions in conjunction with a real-time understanding of mechanisms. Figure 2 highlights the trajectory of innovations. Improving transformer and diffusion models with multimodal inputs, including text and images, facilitates the fast and accurate

transfer of style by leveraging prior knowledge regarding visual differences that yield efficiency as well as higher quality output [21], [24], [27].

V. CONCLUSION

This review traces the development in the techniques for real-time multi-style transfer from early simple methodologies to the more complex neural network methods currently in use. Early techniques—such as edge detection—set the groundwork for more complex methods. Modern approaches, including Convolutional Neural Networks and Adaptive Instance Normalization, offer enormous increases in efficiency, flexibility, and real-time performance. Real-time style transfer has advanced significantly with the use of techniques that make it possible to apply artistic styles to images and videos almost instantaneously. These approaches work fast using pre-trained neural networks for real-time applications such as live streaming and dynamic video editing. Also, some recent works proposed novel frameworks, combining multiple styles capable of handling more complex content variation, by making use of newly introduced concepts, such as the attention mechanism and semantic awareness, for better quality and coherence in transferred styles. Even with these methods emerging, it is also not easy to achieve high performance combined with wide applicability. When summed up, the computational efficiency and stylistic diversity remained elusive for balance within this research field. Much further research is needed to handle these problems and realize real-time multi-style transfer. As such, the technology still proves very useful when it comes to digitizing images and videos, and there is a huge space for further innovation and applications.

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